

Flow matching and Diffusion Models

Lecture 2: Constructing a Training Target

we train the model by minimize a mean squared error:

$$L(\theta) = \|u_t^0(x) - u_t^{\text{target}}(x)\|^2 \quad \text{training target}$$

key terminology: ① Conditional: per single data point

② Marginal: across distribution of data points

Probability path: the path from noise to data.

1) Conditional probability path: $p_t(\cdot|z)$

Dirac distribution: for data point $z \in \mathbb{R}^d$, we denote with δ_z the Dirac delta distribution

sample from δ_z always return z

Conditional probability path ① $p_t(\cdot|z)$ distribution over $\mathbb{R}^d$ means probability distribution

$$p_0(\cdot|z) = p_{\text{init}}, \quad p_1(\cdot|z) = \delta_z$$

Example: Gaussian probability path.

$$p_t(\cdot|z) = N(\alpha_t z, \beta_t^2 \text{Id}) \quad \text{Gaussian conditional path.}$$

α_t, β_t : noise schedules: $\alpha_0 = 0, \alpha_1 = 1; \beta_0 = 1, \beta_1 = 0$ (e.g. $\alpha_t = t, \beta_t = 1-t$)

$$p_{\text{init}} = N(0, \text{Id})$$

$$p_0(\cdot|z) = N(0, \text{Id}) \quad p_1(\cdot|z) = (z, 0) = \delta_z$$

2) Marginal probability path:

$z \sim p_{\text{data}}, X \sim p_t(\cdot|z) \Rightarrow x \sim p_t$ sampling from marginal path.

What is marginal probability path?

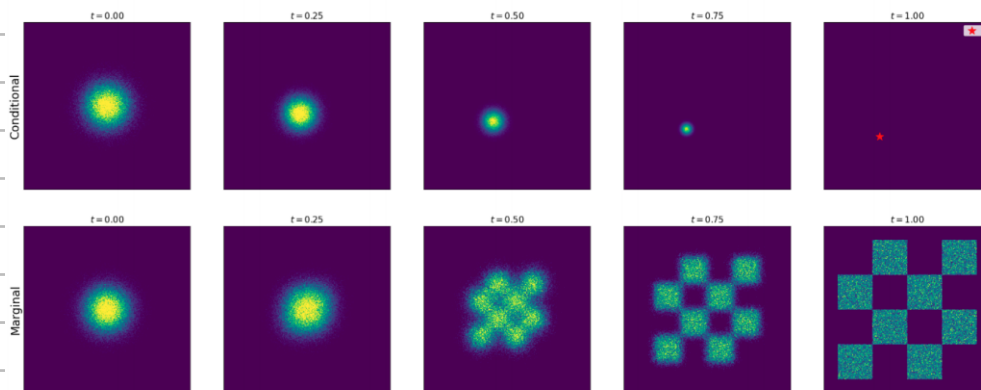
In probability, marginalizing means averaging out or ignoring the effect of one variable to get the distribution of another.

$p_t(x)$ is the result of taking the conditional path and averaging it over all possible z . (consider the outcomes resulting from all possible starting point z).

$$\text{density of marginal path: } p_t(x) = \int p_t(x|z) p_{\text{data}}(z) dz$$

the probability of reaching x from a specific z the probability of z occurring in the first place

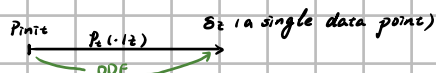
$$\text{② } p_0 = p_{\text{init}}, \quad p_1 = p_{\text{data}}$$



Vector field

1) Conditional vector field: $u_t^{\text{target}}(x|z)$ ($0 \leq t \leq 1, x, z \in \mathbb{R}^d$)

$$x_0 \sim p_{\text{init}}, \quad \frac{d}{dt} x_t = u_t^{\text{target}}(x_t) \Rightarrow x_t \sim p_t(\cdot|z) \quad (0 \leq t \leq 1)$$



if you move according to the VF u_t^{target} , then at any point in time t , your position x_t will perfectly follow the probability path $p_t(\cdot|z)$ that we defined earlier.

Example: Cond Gaussian VF

$$u_t^{\text{target}}(x|z) = \left(\alpha_t - \frac{\beta_t}{\beta_t} \alpha_t \right) z + \frac{\beta_t}{\beta_t} x \quad \left(\alpha_t = \frac{d}{dt} \alpha_t \quad \beta_t = \frac{d}{dt} \beta_t \right)$$

proof: current location: $\psi_t^{\text{target}}(x|z) = \alpha_t z + \beta_t x$

$$\alpha_t z + \beta_t x = \frac{d}{dt} \psi_t^{\text{target}}(x|z) = u_t^{\text{target}}(\psi_t^{\text{target}}(x|z)|z)$$

(x refers to the initial noise X_0 , according to $X_t = \alpha_t z + \beta_t X_0 \Rightarrow X_0 = \frac{X_t - \alpha_t z}{\beta_t}$)

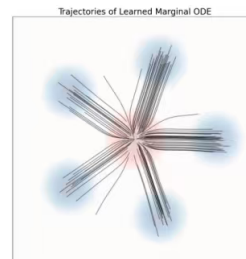
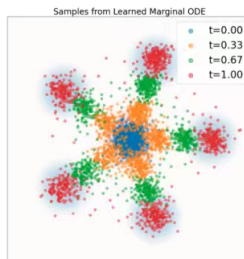
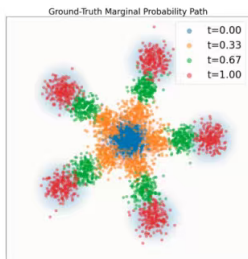
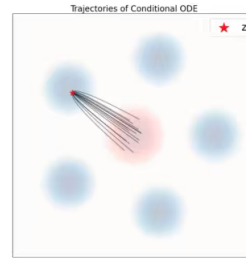
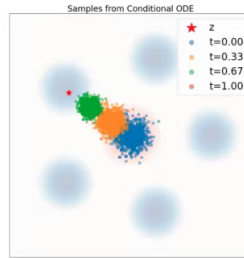
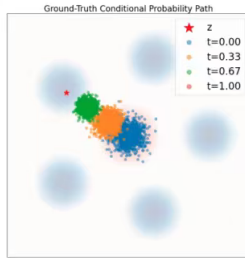
$$\Rightarrow \alpha_t z + \beta_t x = u_t^{\text{target}}(\alpha_t z + \beta_t x|z)$$

$$\Rightarrow \alpha_t z + \beta_t \left(\frac{x - \alpha_t z}{\beta_t} \right) = u_t^{\text{target}}(x|z) \Rightarrow u_t^{\text{target}}(x|z) = \left(\alpha_t - \frac{\beta_t}{\beta_t} \alpha_t \right) z + \frac{\beta_t}{\beta_t} x$$

2) Marginal vector field

$$u_t^{\text{target}}(x) = \int u_t^{\text{target}}(x|z) \frac{p_t(x|z) p_{\text{data}}(z)}{p_t(x)} dz$$

$$X_0 \sim p_{\text{init}}, \quad \frac{d}{dt} X_t = u_t^{\text{target}}(X_t) \Rightarrow X_t \sim p_t \quad (0 \leq t \leq 1) \Rightarrow X_1 \sim p_{\text{data}}$$



Divergence: $\text{div}(v_t)(x) = \sum_{i=1}^d \frac{\partial}{\partial x_i} v_t^i(x)$ $v_t(x)$ is a vector field
the rate of change of v_t in the i -th direction
at the given point x , is the point group diverging outward or converging inward

① $\text{div} > 0$ (source) $\Rightarrow$ density decrease

② $\text{div} < 0$ (sink) $\Rightarrow$ density increase

Continuity Equation: $\partial_t p_t(x) = -\text{div}(p_t u_t^{\text{target}})(x)$

change of probability mass at x outflow - inflow of probability mass from u

$\uparrow \uparrow \uparrow$ outflow
 $\boxed{x}$
 $\uparrow \uparrow \uparrow$ inflow

Score Function

Conditional score function $\nabla \log p_t(x|z)$

Marginal score function $\nabla \log p_t(x)$

$$\begin{aligned} \nabla \log p_t(x) &= \frac{\nabla p_t(x)}{p_t(x)} = \frac{\nabla \int p_t(x|z) p_{\text{data}}(z) dz}{p_t(x)} = \frac{\int \nabla p_t(x|z) p_{\text{data}}(z) dz}{p_t(x)} = \int \nabla \log p_t(x|z) \frac{p_t(x|z) p_{\text{data}}(z)}{p_t(x)} dz \\ &= \int \nabla \log p_t(x|z) p_t(z|x) dz \quad p_t(z|x) = \frac{p_t(x|z) p_{\text{data}}(z)}{p_t(x)} \quad (\text{Bayes Formula}) \end{aligned}$$

the score function of the entire data population $\uparrow$ individual score function

posterior probability: if I am at x now, what's the probability that I came from the starting point z .

Example: Score Function of Gaussian Probability Path.

$$p_t(x|z) = \mathcal{N}(x, \alpha_t z, \beta_t \text{Id})$$

$$\nabla \log p_t(x|z) = -\frac{x - \alpha_t z}{\beta_t}$$

SDE extension trick

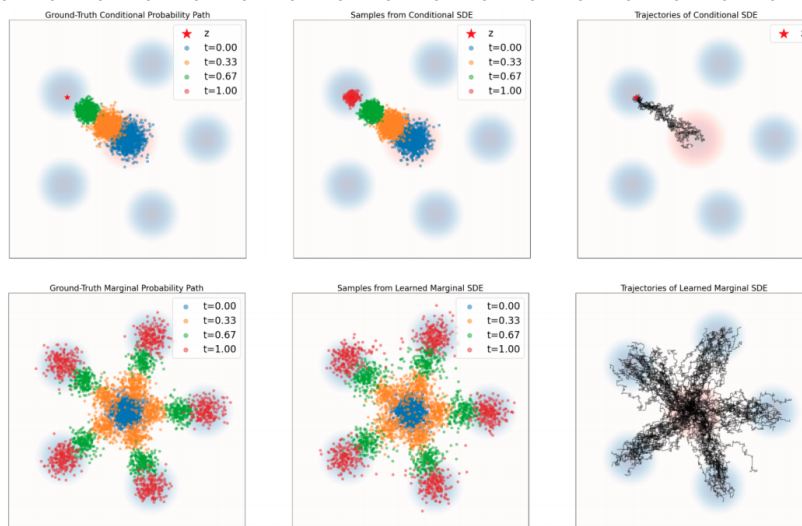
let $u_t^{\text{target}}(x) = \int u_t^{\text{target}}(x|z) \frac{p_t(x|z) p_{\text{data}}(z)}{p_t(x)} dz$ as before.

for any $\delta \geq 0$:

let the probability density p_t tend towards a uniform distribution.

$$X_0 \sim p_{\text{init}}, \quad dX_t = \left[u_t^{\text{target}}(X_t) + \frac{\delta_t}{2} \nabla \log p_t(X_t) \right] dt + \delta_t dW_t \Rightarrow X_t \sim p_t \quad (0 \leq t \leq 1) \Rightarrow X_1 \sim p_{\text{data}}$$

pointing to the area with the highest density combat the random divergence effect of dW_t



Laplacian operator Δ : $\Delta w_t(x) = \sum_{i=1}^d \frac{\partial^2}{\partial x_i^2} w_t(x) = \text{div}(\nabla w_t)(x)$

the overall curvature of function w_t at point x

$\Delta w_t(x) > 0$: ∇ $\Delta w_t(x) > 0$ $\wedge$

Fokker-Planck Equation: $X_0 \sim p_{init}$, $dX_t = u_t(X_t)dt + \sigma_t dW_t$

$\partial_t p_t(x) = -\text{div}(p_t u_t)(x) + \frac{\sigma_t^2}{2} \Delta p_t(x)$

① $\Delta p_t(x) < 0$ ∇ $\partial_t p_t(x) \downarrow$: randomness is pushing the data points outward from this "peak", trying to flatten it.

② $\Delta p_t(x) > 0$ ∇ $\partial_t p_t(x) \uparrow$: randomness is pushing the data points toward this "valley", trying to flatten it.
 $\downarrow$
 simulate noise

$$\begin{aligned} \partial_t p_t(x) &= -\text{div}(p_t u_t^{\text{target}})(x) \\ &= -\text{div}(p_t u_t^{\text{target}})(x) - \frac{\sigma_t^2}{2} \Delta p_t(x) + \frac{\sigma_t^2}{2} \Delta p_t(x) \\ \nabla \log p_t = \frac{\nabla p_t}{p_t} \quad \downarrow &= -\text{div}(p_t u_t^{\text{target}})(x) - \text{div}\left(p_t \left[\frac{\sigma_t^2}{2} \nabla \log p_t\right]\right)(x) + \frac{\sigma_t^2}{2} \Delta p_t(x) \\ &= -\text{div}\left(p_t \left[u_t^{\text{target}} + \frac{\sigma_t^2}{2} \nabla \log p_t\right]\right)(x) + \frac{\sigma_t^2}{2} \Delta p_t(x) \end{aligned}$$

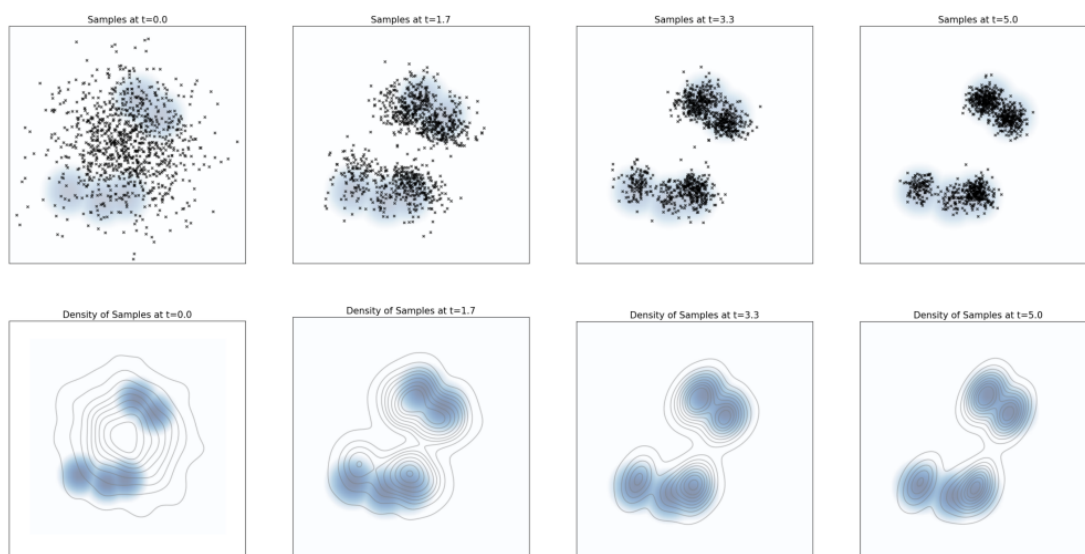


Figure 8: Top row: Particles evolving under the Langevin dynamics given by eq. (31), with $p(x)$ taken to be a Gaussian mixture with 5 modes. Bottom row: A kernel density estimate of the same samples shown in the top row. As one can see, the distribution of samples converges to the equilibrium distribution p (blue background colour).

Langevin dynamics.

when the probability path is static, $p_t = p$ for a fixed distribution p , we set $u_t^{\text{target}} = 0$:

$$dX_t = \left[u_t^{\text{target}}(X_t) + \frac{\sigma_t^2}{2} \nabla \log p_t(X_t) \right] dt + \sigma_t dW_t$$

$\downarrow$

$$dX_t = \frac{\sigma_t^2}{2} \nabla \log p(X_t) dt + \sigma_t dW_t$$

$$\begin{cases} \text{if } X_0 \sim p \Rightarrow X_t \sim p \quad (t \geq 0) \\ \text{if } X_0 \sim p' \neq p \Rightarrow t \rightarrow \infty \quad X_t \sim p \end{cases}$$

Summary

Conditional probability path: $p_t(x|z)$ $p_0(\cdot|z) = p_{\text{init}}$, $p_t(\cdot|z) = \delta_z$

Conditional vector field: $u_t^{\text{flow}}(x|z)$ $X_0 \sim p_{\text{init}} \Rightarrow X_t = \psi_t^{\text{target}}(X_0|z) \sim p_t(\cdot|z)$

Conditional score function $\nabla \log p_t(x|z)$

Marginal probability path $p_t(x)$ $z \sim p_{\text{data}}$, $X \sim p_t(\cdot|z)$, $X \sim p_t$.

Marginal vector field: $u_t^{\text{target}}(x)$ $u_t^{\text{target}}(x) = \int u_t^{\text{target}}(x|z) \frac{p_t(x|z) p_{\text{data}}(z)}{p_t(x)} dz$

Marginal score function $\nabla \log p_t(x)$ $\nabla \log p_t(x) = \int \nabla \log p_t(x|z) \frac{p_t(x|z) p_{\text{data}}(z)}{p_t(x)} dz$

Gaussian probability path $p_t(x|z)$ $p_t(x|z) \sim \mathcal{N}(x, \alpha_t z, \beta_t Id)$

Gaussian vector field: $u_t^{\text{flow}}(x|z)$ $u_t^{\text{flow}}(x|z) = (\alpha_t - \frac{\beta_t}{\beta_t}) z + \frac{\beta_t}{\beta_t} x$

Gaussian score function $\nabla \log p_t(x|z)$ $\nabla \log p_t(x|z) = - \frac{x - \alpha_t z}{\beta_t}$